

Definition. A subgroup H of G is called a *normal subgroup* of G iff:

Theorem (Normal Subgroup Test).

Proof:

Question 1. Let $H = \{(1), (12)(34)\}$ in A_4 . Is H normal in G ?

Question 2. Is A_n a normal subgroup of S_n ? Why or why not?

Question 3. Is $K = \{R_0, H\}$ normal in D_4 ?

Question 4. Suppose H is a subgroup of G and H is the only subgroup of G with order k . Why is H normal in G ?

Exercise 1. Prove that the intersection of two normal subgroups is normal.

Theorem (Factor Groups). Let G be a group and let H be normal in G . Then

Question 5. Let $G = \mathbb{Z}_{12}$. Let $H = \langle 3 \rangle$. What are the elements of G/H ?

Order of Elements in FGs:

Question 6. Consider the group $G = \mathbb{Z}_{36} / \langle 4 \rangle$. What are the orders of the elements in G ?

Question 7. Let $G = \mathbb{Z}_8 \oplus \mathbb{Z}_4$ and let $H = \langle (2, 2) \rangle$. How many elements are in G/H ? Is G/H Abelian? Is G/H cyclic?

Exercise 2. Let G be a finite group and let $H \triangleleft G$. Prove the order of $gH \in G/H$ must divide the order of g in G .

Theorem.

Proof:

Question 8. Let G be a group of order 35 with a non-trivial center. Why must G be Abelian?

Theorem.

Question 9. What can you say about $\text{Inn}(D_6)$?

Theorem (Cauchy's Theorem for Abelian Groups).

Exercise 3. Let G be an Abelian group of order 33. Prove that G is cyclic.

Definition. A group G is the **internal direct product** of subgroups H and K ($G = H \times K$) if:

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The Internal Direct Product of two subgroups is nice because:

Theorem.

Question 10. Why are each of the three conditions of IDP important?

Question 11. How many elements of order 6 are in $\mathbb{Z}_9 \times \mathbb{Z}_6$? How many cyclic subgroups of order 6 are there in this group?

Additional Exercises

1. Determine the order of $(\mathbb{Z} \oplus \mathbb{Z})/\langle(4, 2)\rangle$. Is this group cyclic?
2. In $\mathbb{Z}$, let $H = \langle 5 \rangle$ and $K = \langle 7 \rangle$. Does $\mathbb{Z} = HK$? Does $\mathbb{Z} = H \times K$?
3. Let p be a prime and G be a group of order $2p$. Show that if H is a subgroup of G that is not normal, then H has order 2.

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